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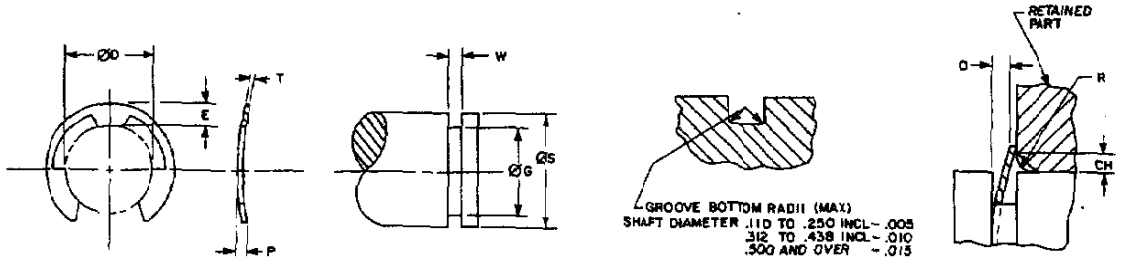
PROCUREMENT SPECIFICATION
MIL-R-21248

TITLE
RING, RETAINING, EXTERNAL, BOWED "E" (REDUCED SECTION TYPE)

MILITARY STANDARD
MS16634

PAGE 1 OF 5

Ⓑ ENTIRE STANDARD REVISED



ENLARGED DETAILS

TABLE I. DIMENSIONS

ØS SHAFT (REF)		ØD FREE		E LARGE SECTION HEIGHT	T 1/ THICKNESS		P OVERALL BOW HEIGHT		ØG RECOMMENDED GROOVE (REF)		W WIDTH		O 3/ TAKE UP	R 4/ OF RETAINED PART (REF)	CH 4/
INCH	MM	BASIC	TOL	(REF)	BASIC	TOL	BASIC	TOL	BASIC	TOL	BASIC	TOL	MAX	MAX	MAX
.110	2.8	.075		.155	.010		.030	±.005	.079	+.002 -.000	.035		.013	.080	.060
.125	3.2	.094		.073	.010	±.001	.030		.096	.0015 FIM 2/	.022		.013	.040	.030
.140	3.6	.102	+.002	.090	.015		.033		.105		.025		.014	.040	.045
.156	4.0	.114	-.003	.090	.015		.035		.116		.027		.014	.060	.045
.172	4.4	.125		.099	.015		.037	±.005	.127	+.003 -.000 -.003 FIM 2/	.029		.016	.060	.045
.188	4.8	.145		.101	.015		.038		.147		.030		.027	.060	.045
.219	5.6	.185		.132	.015		.051		.188		.040		.026	.060	.045
.250	6.4	.207		.167	.025		.058	+.007	.210		.049	+.005	.023	.060	.045
.312	7.9	.243		.136	.025		.058	-.008	.250		.047	-.000	.023	.060	.045
.375	9.5	.300	+.003	.188	.035		.068	±.008	.303		.060		.022	.065	.050
.438	11.1	.337	-.004	.183	.035	±.002	.068		.343		.060		.022	.065	.050
.500	12.7	.392		.213	.042		.084	±.009	.396	+.004 -.000 .004 FIM 2/	.073		.030	.080	.060
.625	15.9	.480		.240	.042		.089		.485		.077		.035	.080	.060
.744	18.9	.616		.202	.050		.100		.625		.085		.037	.060	.045
.750	19.0	.574	+.003 -.005	.283	.050		.100	±.010	.580		.085		.037	.085	.065
.875	22.2	.668		.326	.050		.100		.675		.085		.037	.085	.065
.984	25.0	.822		.349	.050		.100		.835		.085		.037	.085	.065
1.188	30.2	1.066	+.006	.297	.062	±.003	.124		1.079	+.005 -.000 .005 FIM 2/	.107		.045	.090	.070
1.375	34.9	1.213	-.010	.348	.062		.124	±.015	1.230		.107		.045	.090	.070

1/ = THICKNESS "T" APPLIES TO UNPLATED RINGS. FOR CORROSION RESISTANT STEEL AND PLATED RINGS, +.002 SHOULD BE ADDED TO THE MAXIMUM TOLERANCE, i.e. ±.002 SHOULD BE +.004/-.002.
2/ FIM = (FULL INDICATOR MOVEMENT) IS THE MAXIMUM ALLOWABLE DEVIATION OF CONCENTRICITY BETWEEN THE GROOVE AND THE SHAFT.
3/ O = END PLAY TAKE-UP (DESIGN REFERENCE DIMENSION).
4/ R AND CH = RADII OR CHAMFERS ALLOWABLE ON PARTS TO BE RETAINED BY THE RINGS. THRUST LOADS OF RINGS, RETAINING PARTS WITH CORNER RADII OR CHAMFERS, SEE Note (h) ON PAGE 5.

APPROVED 11 DEC 1958 REVISED Ⓑ 2 MAY 1990

FED SUP CLASS 5365

5365 - 0126

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REQUIREMENTS:

1. CLASSIFICATION: RETAINING RINGS FURNISHED UNDER THIS STANDARD SHALL BE TYPE II, CLASS 3 OF THE PROCUREMENT SPECIFICATION.

2. MATERIAL:

(a) CARBON SPRING STEEL, GRADE 1060 THRU 1095 (UNS G10600 THRU G10950) IN ACCORDANCE WITH ASTM A568 OR ASTM A682.

(b) CORROSION RESISTANT STEEL IN ACCORDANCE WITH AMS 5520 (UNS S15700).

(c) BERYLLIUM COPPER ALLOY NUMBER 170 (UNS C17000) OR ALLOY NUMBER 172 (UNS C17200) IN ACCORDANCE WITH ASTM B194.

3. HARDNESS:

TABLE II. HARDNESS

Ø SHAFT (REF)	CARBON STEEL	CORROSION RESISTANT STEEL	BERYLLIUM COPPER
.110 TO .219 INCL	84.5-87HR15N	82.5-86HR15N	77-82HR15N
.250 TO .312 INCL	66.5-71HR30N	63-69.5HR30N	77-82HR15N
.375 TO 1.375 INCL	47-52HRC	44-51HRC	-

4. PROTECTIVE FINISH OR SURFACE TREATMENT:

(a) CARBON STEEL - SHALL BE AS SPECIFIED (SEE TABLE III OR IV):

(1) CADMIUM PLATE IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 3 OR ASTM B696, TYPE II, CLASS 5.

(2) ZINC COAT IN ACCORDANCE WITH ASTM B633, TYPE II, CLASS Fe/Zn5, OR ASTM B695, TYPE II, CLASS 5.

(3) PHOSPHATE COAT IN ACCORDANCE WITH DOD-P-16232, TYPE Z, CLASS 2.

(b) CORROSION RESISTANT STEEL - SHALL BE CLEANED, DESCALED AND PASSIVATED IN ACCORDANCE WITH QQ-P-35.

5. PART NUMBER: THE BASIC MS PART NUMBER IS FOLLOWED BY A DASH NUMBER TAKEN FROM TABLE III OR IV.

EXAMPLE: MS16634-1011 IS THE PART NUMBER FOR A CARBON STEEL CADMIUM PLATE, EXTERNAL, BOWED "E" RETAINING RING FOR USE ON A .110 DIAMETER SHAFT.

NOTES:

1. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.

2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.

3. REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.

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P.A.	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
NAVY - OS Other Cust ARMY - AR AIRFORCE - 99		RING, RETAINING, EXTERNAL, BOWED "E" (REDUCED SECTION TYPE)	MS16634
PROCUREMENT SPECIFICATION: MIL-R-21248	SUPERSEDES		PAGE 2 OF 5

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TABLE III. DASH NUMBERS FOR MS16634

Ø S SHAFT (REF)	CARBON STEEL 1/ CADMIUM PLATE	CARBON STEEL 1/ ZINC COAT	CARBON STEEL 1/ PHOSPHATE COAT	CORROSION RESISTANT STEEL	BERYLLIUM 1/ COPPER
	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.
.110	-1011	-2011	-3011	-4011	-5011
.125	-1012	-2012	-3012	-4012	-5012
.140	-1014	-2014	-3014	-4014	-5014
.156	-1015	-2015	-3015	-4015	-5015
.172	-1017	-2017	-3017	-4017	-5017
.188	-1018	-2018	-3018	-4018	-5018
.219	-1021	-2021	-3021	-4021	-5021
.250	-1025	-2025	-3025	-4025	-5025
.312	-1031	-2031	-3031	-4031	-5031
.375	-1037	-2037	-3037	-4037	-5037
.437	-1043	-2043	-3043	-4043	-5043
.500	-1050	-2050	-3050	-4050	-5050
.625	-1062	-2062	-3062	-4062	-5062
.744	-1074	-2074	-3074	-4074	-5074
.750	-1075	-2075	-3075	-4075	-5075
.875	-1087	-2087	-3087	-4087	-5087
.984	-1098	-2098	-3098	-4098	-5098
1.188	-1118	-2118	-3118	-4118	-5118
1.375	-1137	-2137	-3137	-4137	-5137

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY, OR IN FUEL OR LUBRICATION SYSTEMS, OR WHEN USED AT TEMPERATURES OVER 450°F (233°C).

TABLE IV. SUBSTITUTION TABLE (CROSS REFERENCE OF PART NUMBERS)

Ø S SHAFT (REF)	INACTIVE	SUBSTITUTE	SUBSTITUTE	SUBSTITUTE
	CARBON STEEL	CARBON 1/ STEEL CADMIUM PLATE	CARBON 1/ STEEL ZINC COAT	CARBON 1/ STEEL PHOSPHATE COAT
	MS16634	MS16634	MS16634	MS16634
.110	-11	-1011	-2011	-3011
.125	-12	-1012	-2012	-3012
.140	-14	-1014	-2014	-3014
.156	-15	-1015	-2015	-3015
.172	-17	-1017	-2017	-3017
.188	-18	-1018	-2018	-3018
.219	-21	-1021	-2021	-3021
.250	-25	-1025	-2025	-3025
.312	-31	-1031	-2031	-3031
.375	-37	-1037	-2037	-3037
.437	-43	-1043	-2043	-3043
.500	-50	-1050	-2050	-3050
.625	-62	-1062	-2062	-3062
.744	-74	-1074	-2074	-3074
.750	-75	-1075	-2075	-3075
.875	-87	-1087	-2087	-3087
.984	-98	-1098	-2090	-3090
1.188	-118	-1118	-2118	-3118
1.375	-137	-1137	-2137	-3137

1/ SUBSTITUTE CORROSION RESISTANT STEEL WHEN USED IN FOOD PROCESSING MACHINERY, OR IN FUEL OR LUBRICATION SYSTEMS, OR WHEN USED AT TEMPERATURES OVER 450°F (233°C).

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P.A.	INTERNATIONAL INTEREST	TITLE	MILITARY STANDARD
NAVY - OS Other Cust ARMY - AR AIRFORCE - 99		RING, RETAINING, EXTERNAL, BOWED "E" (REDUCED SECTION TYPE)	MS16634
PROCUREMENT SPECIFICATION MIL-R-21248	SUPERSEDES:		PAGE 3 OF 5

DD FORM 1 MAY 72 672-1 (COORDINATED) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE 5365 - 0126

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RECOMMENDED DESIGN LIMITATIONS AND USAGE

(a) INTENDED USE - TO PROVIDE UNUSUALLY LARGE SHOULDERS FOR POSITIONING AND MAINTAINING MACHINE COMPONENTS ON SHAFTS. THEY ARE APPLIED RADIALLY AND WITHSTAND CONSIDERABLE THRUST LOAD. THE RINGS ARE BENT LIKE A BOW OUT OF PLANE. FREE ENDS AND OPPOSITE EDGE ABUT THE MACHINE PART, MID SECTION OF RING ABUTS OUTER GROOVE WALL. RINGS WILL TAKE UP END PLAY RESILIENTLY. THEY ARE OF ADVANTAGE WHERE AXIAL ASSEMBLY OF A BOWED RETAINING RING IS NOT POSSIBLE. THE USE OF THE FOLLOWING FORMULAS ARE BASED ON THE FACT THAT THE RING MATERIAL WILL NOT FAIL IN COMPRESSION.

LIMITATION ON USE - THE FOLLOWING FORMULAS ARE NOT TO BE USED FOR BRITTLE MATERIALS SUCH AS CAST IRON, ETC.

WARNING - RINGS SHOULD NOT BE OVER EXPANDED DURING INSTALLATION SINCE THIS WILL LEAD TO RING FAILURE. IF RING HAS PLAY BETWEEN THE GROOVE DIAMETER AND THE INSIDE RING DIAMETER THIS INDICATES THAT THE RING HAS BEEN OVER EXPANDED, (PROVIDING GROOVE WAS BEEN MACHINED TO RECOMMENDED DIMENSIONS).

FOR APPROXIMATE SAFETY RPM LIMITS SEE TABLE V.

TABLE V. APPROXIMATE SAFETY RPM LIMITS

Ø SHAFT (INCHES)		.250	.500	1.000
CARBON STEEL AND CORROSION RESISTANT STEEL	RPM LIMIT	25,000	14,000	6,500
BERYLLIUM COPPER	RPM LIMIT	16,000	9,000	-

(b) ALLOWABLE THRUST LOAD CAPACITY OF THE RING (ABUTTING COMPONENTS TO HAVE SHARP CORNERS) =

$$P = \frac{\pi S T X}{3F}$$

WHERE:
P = ALLOWABLE THRUST LOAD (POUNDS)
S = SHAFT DIAMETER (INCHES)
T = RING THICKNESS (INCHES)
X = ULTIMATE SHEAR STRENGTH OF THE RING MATERIAL (PSI) 1/
F = FACTOR OF SAFETY

A SAFETY FACTOR, F=3, IS RECOMMENDED SINCE THE RING UNDER LOAD IS SUBJECTED NOT ONLY TO PURE SHEAR STRESSES BUT ALSO TO BENDING STRESSES.

(c) ALLOWABLE LOAD CAPACITY OF GROOVE WALL =

$$P = \frac{\pi G d Y}{3F}$$

WHERE:
P = ALLOWABLE COMPRESSION LOAD (POUNDS)
G = GROOVE DIAMETER (INCHES)
d = GROOVE DEPTH (INCHES)
Y = YIELD STRENGTH IN COMPRESSION OF THE GROOVE MATERIAL (PSI)
F = FACTOR OF SAFETY

TO INSURE A SAFE WORKING LOAD, A SAFETY FACTOR, F = 2 IS RECOMMENDED.

(d) MINIMUM DISTANCE BETWEEN OUTER GROOVE WALL AND END OF SHAFT =

$$Z = 2d$$

WHERE:
Z = MINIMUM DISTANCE BETWEEN OUTER GROOVE WALL AND END OF SHAFT (INCHES)
d = GROOVE DEPTH (INCHES)

1/ X = 100,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS UP TO AND INCLUDING .219 INCH SHAFT DIAMETER OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 120,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS .250 AND .312 INCH SHAFT DIAMETER OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 150,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS .375 INCH AND OVER SHAFT DIAMETER OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 110,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS OF ALL SHAFT DIAMETERS AND OF BERYLLIUM COPPER.

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(e) LOCATION OF GROOVE =

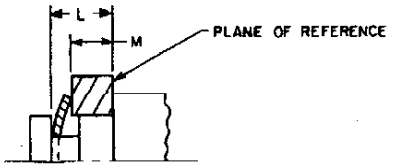
$L_{MAX} = M_{MIN} + T_{MIN}$ + TAKE-UP (AS LISTED)

$L_{MIN} = M_{MAX} + T_{MAX} + .005$ 1/

TAKE-UP $\geq \Delta_L + \Delta_M + \Delta_T + .005$ WHERE: $\Delta_L = L_{MAX} - L_{MIN}$

$\Delta_M = M_{MAX} - M_{MIN}$

$\Delta_T = T_{MAX} - T_{MIN}$



(f) VIBRATION =

A RETAINING RING WILL BE SECURE AGAINST VIBRATION IF ITS SHEAR FATIGUE STRENGTH EQUALS OR EXCEEDS THE FORCE CREATED BY THE MASS OF THE PART BEING REPEATEDLY ACCELERATED AGAINST THE RING.

(1) SHEAR STRENGTH OF RING (CRITICAL) (ABUTTING COMPONENTS TO HAVE SHARP CORNERS).

$.36XTS \geq \frac{W}{g} a$ WHERE:

OR FOR HARMONIC MOTION $a = \delta\omega^2$

$.36XTS \geq \frac{W}{g} \delta\omega^2$

X = ULTIMATE SHEAR STRENGTH OF RING MATERIAL (PSI) 2/
T = RING THICKNESS (INCHES)
S = SHAFT DIAMETER (INCHES)
W = WEIGHT OF ABUTTING PARTS (POUNDS)
g = ACCELERATION DUE TO GRAVITY (IN/SEC^2)
δ = AMPLITUDE OF VIBRATION (INCHES)
ω = ANGULAR SPEED (RAD/SEC)
a = ACCELERATION (IN/SEC^2)

(2) COMPRESSION STRENGTH OF GROOVE MATERIAL (CRITICAL).

$.55YdG \geq \frac{W}{g} \delta\omega^2$ WHERE:

Y = YIELD STRENGTH IN COMPRESSION OF THE GROOVE MATERIAL (PSI)
d = GROOVE DEPTH (INCHES)
G = GROOVE DIAMETER (INCHES)
OTHER SYMBOLS AS SHOWN IN (1)

(g) IMPACT CAPACITY OF RING OR GROOVE WALL =

$I_R = \frac{PT}{2}$ - FOR THE RING (INCH POUNDS) (ABUTTING COMPONENTS TO HAVE SHARP CORNERS)

$I_G = \frac{Pd}{2}$ - FOR THE GROOVE (INCH POUNDS)

WHERE:
P = ALLOWABLE THRUST LOAD OF RING OR GROOVE (POUNDS)
T = RING THICKNESS (INCHES)
I_G = IMPACT CAPACITY OF GROOVE WALL (INCH POUNDS)
d = GROOVE DEPTH (INCHES)
I_R = IMPACT CAPACITY OF RING (INCH POUNDS)

(h) LOAD CAPACITY, WITH THE RETAINED PART RADIUSED OR CHAMFERED.
THE ALLOWABLE THRUST LOADS, WHEN COMPUTED FROM THE FORMULAS ON PAGE 4, WILL NOT BE AFFECTED WHEN THE (E) RINGS ARE USED AGAINST PARTS HAVING CORNER RADII OR CHAMFERS UP TO THE LISTED MAXIMUM DIMENSIONS.

1/ IT IS ADVISABLE FOR SAFER RING ASSEMBLY TO CALCULATE L_{MIN} WITH AN ADDITIONAL .005 IN ORDER THAT THE RING IS NOT FLATTENED OUT COMPLETELY IN ADVERSE TOLERANCE CASES.

2/ X = 100,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS UP TO AND INCLUDING .219 INCH SHAFT DIAMETER OF CARBON STEEL OR CORROSION RESISTANT STEEL.
X = 120,000 PSI ULTIMATE SHEAR STRENGTH FOR RINGS .250 AND .312 INCH SHAFT DIAMETER OF CARBON STEEL OR CORROSION RESISTANT STEEL.
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